

Lab 2 – TrashTag Product Specification

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1 Introduction

1.1 Purpose

1.2 Scope

TrashTag is a browser application designed to improve the waste cleanup process by allowing users to photograph and report litter and organize and conduct cleanups. Live reporting will be available for users to get real-time litter reports and users will be alerted to nearby litter. Images of litter will be provided to users as well as verification of whether the litter is still there or not. A ranking system will be used as gamification for the cleanup process. Users will have access to a leaderboard to see where they rank against other users in their area or even worldwide.

TrashTag allows users to take pictures of trash, pin its location on a map, upload it to the website, and organize clean up groups to help dispose of illegally dumped litter. Local governments will spend less money searching for litter when people in the community can tag it for them, and volunteers will have an easier time organizing events when they are all using the same application. When local governments, non-profits, and volunteers are all connected in one browser application, they can more easily work together to clean up litter.

TrashTag aims to help the environment and the public by providing a resource that clearly highlights the location and type of waste as well as providing live updates regarding existing cleanup efforts and geographic littering patterns to help volunteers allocate their time and resources effectively.

1.3 Definitions, Acronyms, and Abbreviations

API: Application Programming Interface. A software mechanism for accepting and providing data from and to external applications

Illegal Dumping: The unauthorized disposal of trash and other hazardous materials onto private and/or public property.

Metadata: Descriptive information within data files such as date, location, and timestamp data in photos to be submitted to TrashTag.

Geotagging: The addition of GPS coordinates to images that enable precise location identification where trash was reported.

Live Real-Time Updates: Live data synchronization displaying trash reports to both users and cleanup organizations instantaneously.

Gamification: The incorporation of game-like mechanics such as rankings to motivate user participation and enhance cleanup operations.

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1.5 Overview

2 Overall Description

2.1 Product Perspective

2.2 Product Functions

2.3 User Characteristics

2.4 Constraints

2.5 Assumptions and Dependencies